

The logo for Svan, featuring the word "svan" in a bold, blue, sans-serif font. The background of the cover has blue wavy lines at the top and bottom.

svan

SERVICE MANUAL

SGW4600X

Service Manual

For Built-in gas hob

Model list

30 Series Model

60 Series Model

75 Series Model

90 Series Model



Warnings:

- This service manual is not available to the general users, but only applies to professional technicians with relevant qualifications. Any incorrect maintenance may result in other dangers!
- The power and gas supply must be cut off before maintenance!
- A new power cord which meets the relevant technical requirements must be used once possible hidden malfunctions are found with the current power cord (refer to this Manual)!
- Please read carefully all safety warnings in this Manual before maintenance!
- The product is prohibited from using until test is conducted as per relevant regulations after maintenance!
- This service manual applies to many models, so the specific operations please prevail in kind!

Table of contents

1. Precaution	3
1.1 Safty Precaution	3
1.2 Warning	3
2. Dimension	3
3. Description of the appliance	4
3.1. Features	4
4. Electric diagram	4
5. Installation	5
6. Reassembly and Disassembly	6
6.1. Safty precautions	6
6.2. Tools	6
6.3. Reassembly	7

<u>6.3.1. Reassemble of the Ignition Pin</u>	7
<u>6.3.2. Reassemble of the Thermocouple</u>	7
<u>6.3.3. Reassemble of the Gas Pipe</u>	7
<u>6.3.4 Reassemble of Terminal Box</u>	7
<u>6.3.5 Reassemble of Gas Valves</u>	8
<u>6.4.Disassembly</u>	8
<u>6.4.1. Disassemble of the Top Plate</u>	8
<u>6.4.2. Disassemble of the Thermocouple</u>	9
<u>6.4.3. Disassemble of the Ignition Pin</u>	10
<u>6.4.4. Disassemble of the Microswitch Harness</u>	10
<u>6.4.5. Disassemble of the Gas Pipe</u>	10
<u>6.4.6. Disassemble of the Bottom Cup</u>	11
<u>6.4.7. Disassemble of the Valve & Manifold Pipe Assembly</u>	11
<u>6.4.8.Disassemble of the Gas Connection</u>	11
<u>6.5.Check point after assemble</u>	12
<u>7. Troubleshooting</u>	13
<u>7.1.Troubleshooting</u>	13
<u>7.2.Problems and corrective measures</u>	14
<u>7.3.Maintenance and test</u>	15

1. Precaution

1.1. Safety Precaution

1. Do not modify this appliance.
2. This appliance should be serviced by an authorized person.
3. All repairs should be done in accordance with the procedures described in this manual.
4. Before you begin, turn off the gas and electricity supply to the appliance.
5. Remove remaining gas in product completely by igniting burner.
6. Be sure that all of built-in protective devices are replaced. Restore any missing protective shields.
- 7 Do not connect the gas and electricity supply before replacement complete.
8. Use replacement components that have the same ratings. A replacement part that does not have the same safety characteristics as the original might create shock, fire or other hazards
9. You must conduct a gas leakage test after receiving after-sales service in gas pipe and gas connector.
10. Never touch any circuit wiring with your hand nor with uninsulated tool during operation.

1.2 Warning



1. All apparatus within a radius of 50mm from this product must resist temperature of greater than 75°C, otherwise, deformations are easily to be caused during the application process of this product.
2. When the wrapping materials are removed, keep such materials like metal sheet, packing bag, foam and screws out of reach of children's reach to avoid potential dangers. For example, the children may suffocate as a result of swallowing tiny components or playing with packing bag.

2. Dimension (Follow the corresponding order&instructions manual)



60 Series Model



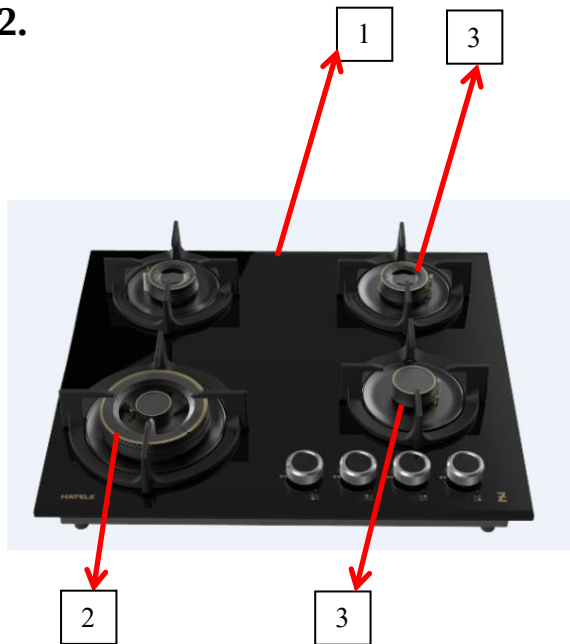
Note: Dimensions for reference only

Description of the appliance(Follow the corresponding order&instructions manual)

3.

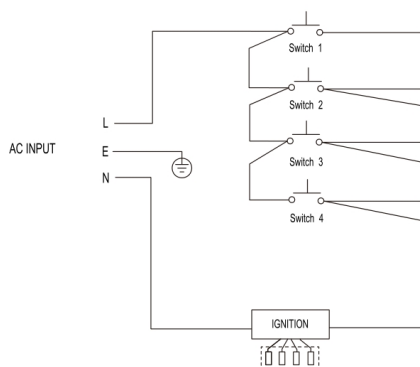
3.1. Features

3.2.

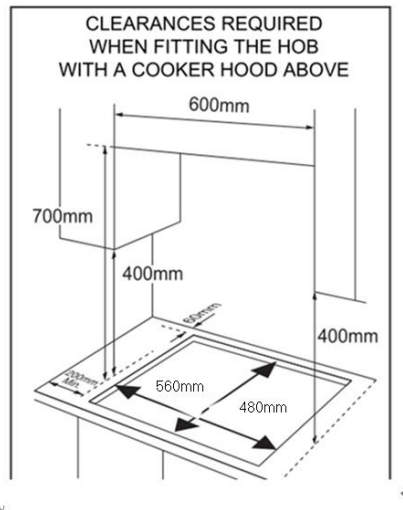


1. Top-plate
2. Triple-crown burner Knob
3. semi rapid burner
4. Auxiliary burner

4. Electric Diagram (Follow the corresponding order&instructions manual)



5.Installation(Follow the corresponding order&instructions manual)



This appliance is to be built into a kitchen unit or worktop of right size, providing the following minimum distances are allowed;

–FH-GS6020 SVSS/FH-GS6020 SVGL:

If the size of A is 340mm, the smallest size of B is 120mm.

If the size of A is 420mm, the smallest size of B is 60mm.

FH-GS6030 SVSS/FH-GS6030 SVGL:

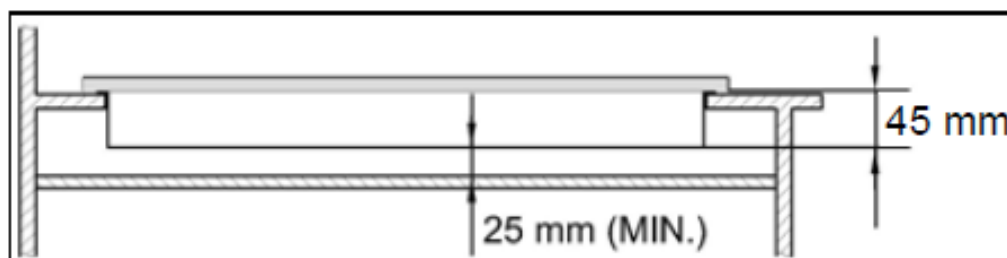
If the size of A is 385mm, the smallest size of B is 120mm.

If the size of A is 470mm, the smallest size of B is 60mm.

– A minimum distance of 200mm must be left between the side edges of the appliance and any adjacent cabinets or walls.

– The minimum distance combustible material can be fitted above the appliance in line with the edges of the appliance is 400mm.

– The minimum distance combustible material can be fitted directly above the appliance is 800mm.



–Gas pipe or connector shouldn't be bent or blocked by any other appliances

– You must conduct a gas leakage test after installation over gas connection and gas pipe.

– An oven must have forced ventilation to install a hob above it.

– Check the dimensions of the oven in the installation manual.

– The cut out size must obey the indication.

6. Reassembly and Disassembly

6.1. Safety precautions



WARNING

Turn off power for safety and appliance protection.
Close middle valve to prevent gas leakage
Keep away from inflammable materials around appliance.
Before work, put on gloves.



CAUTION

Cool off the appliance for a while right after using appliance for your safety because you can get burners by high temperature from the appliances.

6.2. Tools

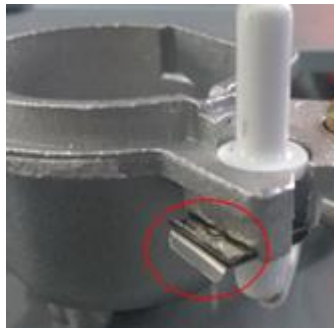
Most of the tools that you might need are shown below. Some are optional.

Item	How to use	Pictures
Screw driver	Use for assembly and disassembly of all screws	
Tubing Wrench	Use for assembly and disassembly of tubing to the burner cup	
7mm Vox Driver	Use for assembly and disassembly of injector nozzles.	
7mm spanner	Use for assembly and disassembly of injector nozzles. (Triple burner)	

6.3. Reassembly

6.3.1 Reassemble of the Ignition Pin

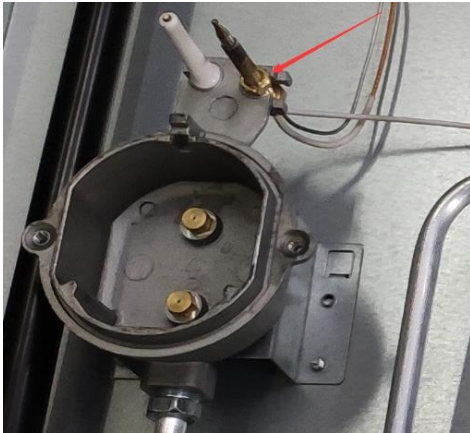
- Step 1. Pull the line of the ignition pin in to the hole on bottom cup along the gap on bottom cup
- Step 2. Push ignition pin to adjust the position
- Step 3. Klemm ignition pin with snap spring to fox the ignition pin.



6.3.2 Reassemble of the Thermocouple

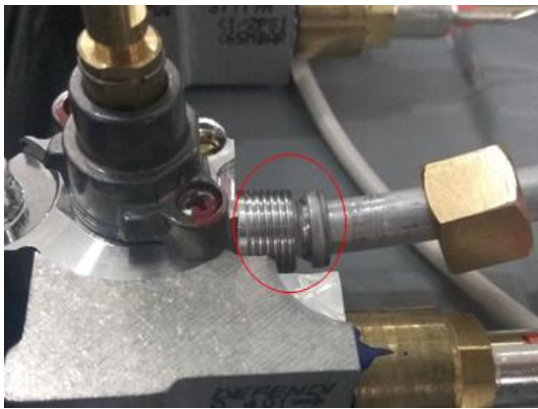
- Step 1. Insert thermocouple into the the relevant hole of the bottom cup from bottom to top.
- Step 2. Adjust position of the thermocouple in the hole of the bottom cup.

Step 3. screw up the nut to fix the thermocouple.



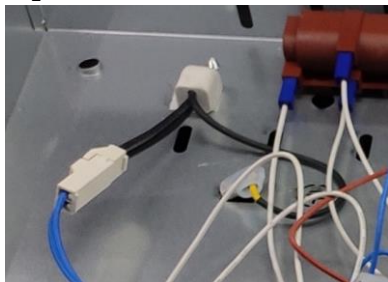
6.3.3 Reassemble of the Gas Pipe

Step 1. Insert the head of the gas pipe into installation parts of the bottom cup or the valve
Step 2. Screw up the nut of the joint with spanner



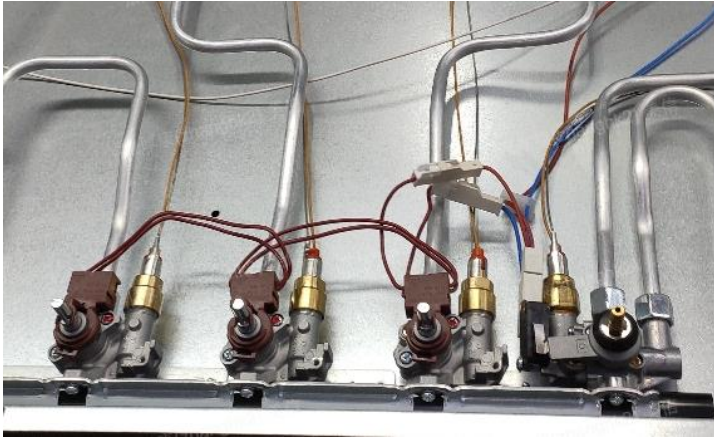
6.3.4 Reassemble of Power line

Step 1. Remove the earthing terminal
Step 2. Remove connector



6.3.5 Reassemble of Gas Valves

Step 1. Place valves in correct position.(refer to the table 6.3.5 below)
Step 2. Lock valves to the bracket with screws.
Step 3. Perform air leakage test(Refer to 7.3 Maintenance and test)

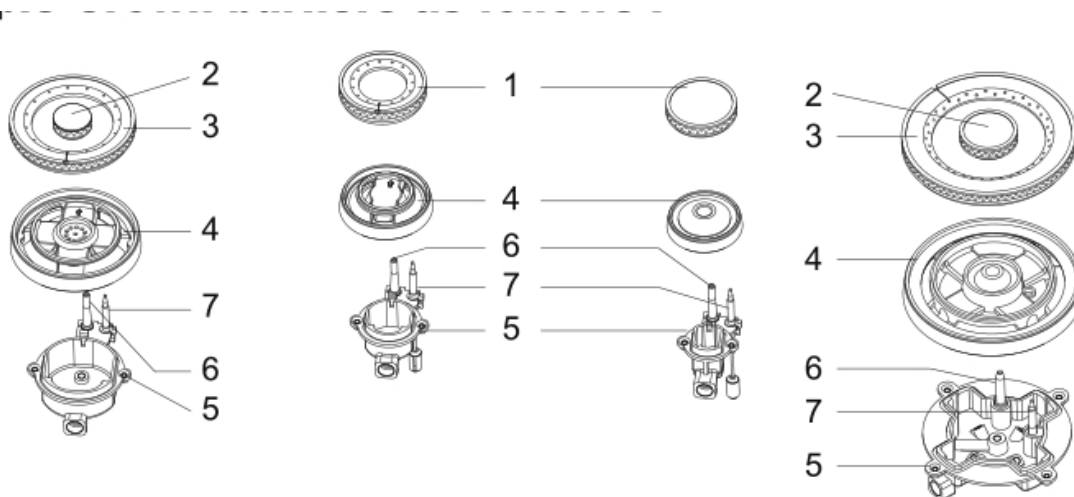


6.4. Disassembly

6.4.1 Disassemble the Top Plate

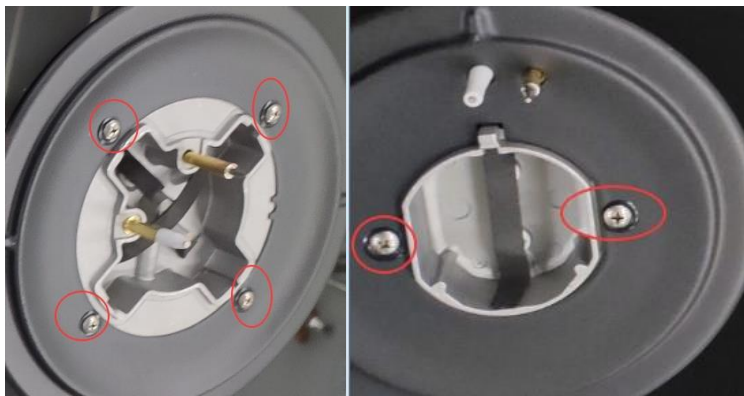
Step 1: Remove parts on the Top Plate

- Remove parts on the top plate. → Burner lids(1/2/3), Flame spreaders(4), Knobs
- When disassembling, keep parts safe, do not drop them to the appliance.



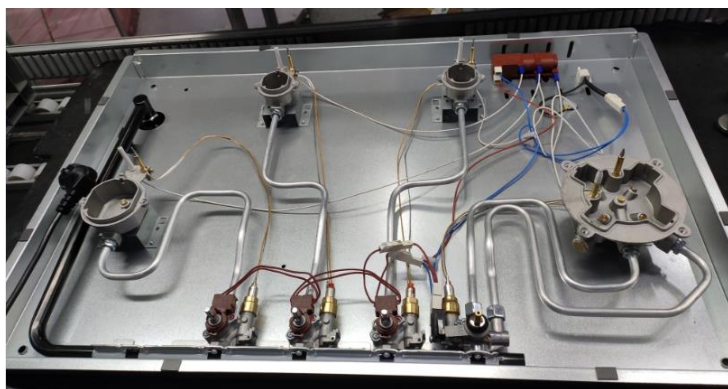
Step 2: Disassemble of Screws on the Top Plate

- Remove the screws of the burner
- Be careful in order not to damage screw heads when disassembling.
- Be careful not to damage the top plate.



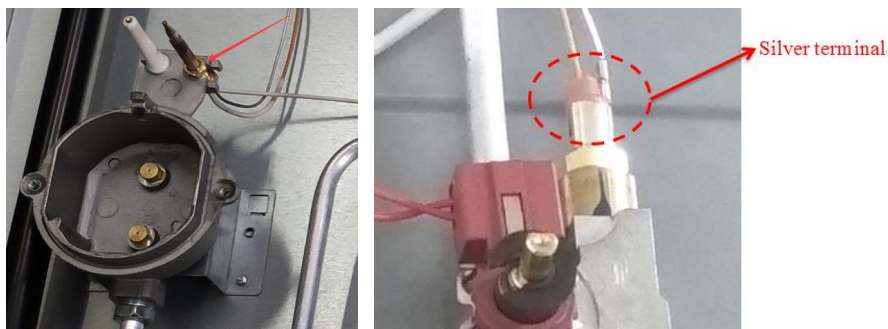
Step 3: Detachment of the Top Plate

- Separate the top plate. (In case of stainless steel model, take off the sponge.)
- During assembly, be careful in order not to give physical impact on parts such as ignition devices, flame supervision devices..



6.4.2 Disassemble the Thermocouple

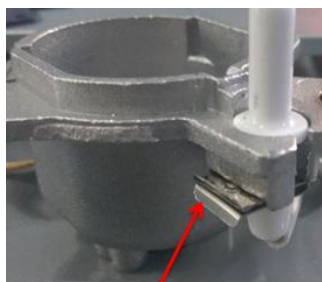
- Step 1: Pull out the snap spring or the nut.
- Step 2: Pull the silver terminal of Thermocouple
- Step 3: When reassembling, replace parts correctly



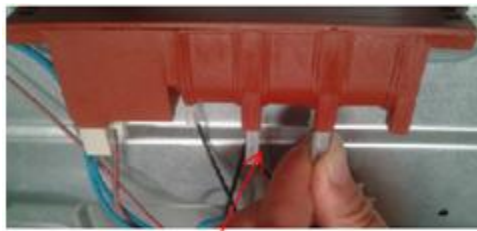
6.4.3 Disassemble of the Ignition Pin

- Step 1: Pull out the snap spring and take off ignition pin.

Step 2: Pull the housing of Ignition pin out of the igniter.



Snap spring



ignition

6.4.4 Disassemble of the Microswitch Harness

Step 1: Before you disassemble the harness, take off the plastic e – ring.

Step 2: When you take off the micro switch harness , please be careful not to break the hook.

Step 3: When you reassemble the harness, replace it correctly.

Step 4: Detach the wire from the power cord assembly harness.

Step 5: Hold housing of the wire by hand and pull it out of the igniter



e-ring



Hook

6.4.5 Disassemble of the Gas Pipe

Step 1: Unscrew the nuts out. at both ends of the pipes

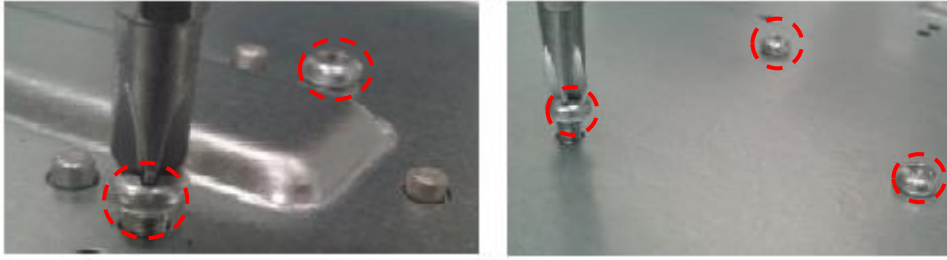
Step 2: Pull the pipes



6.4.6 Disassemble of the Bottom Cup

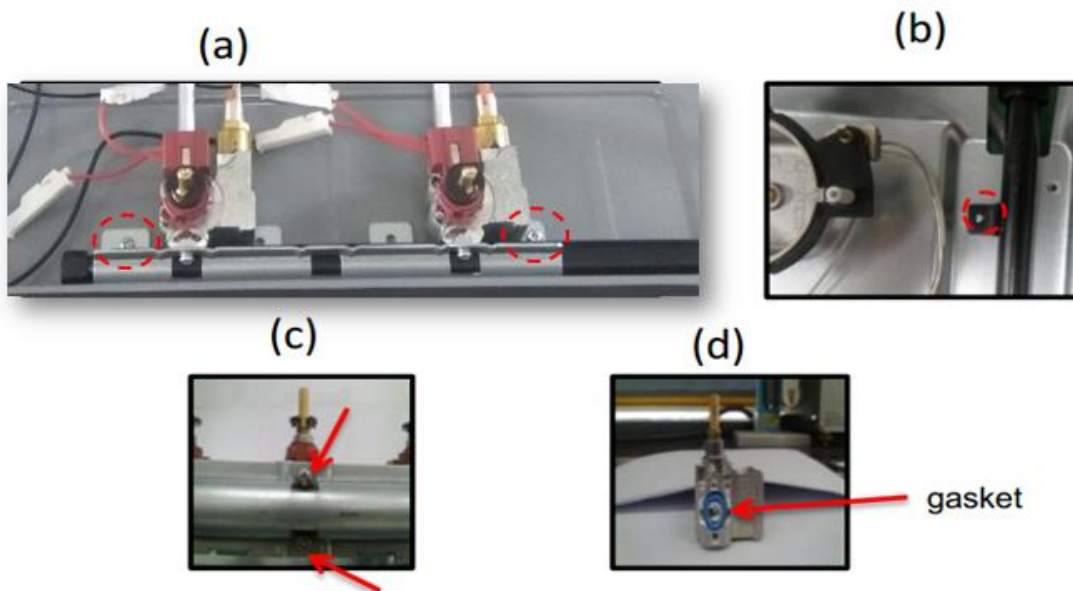
Step 1: Remove all screws on the bottom cup

Step 2: Detach the bottom cup from the base box burner



6.4.7 Disassemble of the Valve & Manifold Pipe Assembly

- Step 1. Remove 2 screws
- Step 2. Remove the screw to take off pipe assy.
- Step 3. Remove 2 screws and detach the valve from the manifold pipe assembly
- Step 4. You must check the gasket when you disassemble or assemble.
- Step 5. If you have faulty one, replace it with new one.
- Step 6. Conduct a leakage test after reassembling



6.4.8 Disasmble of the Gas Connection

- a. When you disassemble/assemble gas connection, use 2 spanners. One for fixing gas connection elbow, the other for tightening gas hose.
- b. Tighten enough to prevent gas leakage.
- c. Once you're done, conduct leakage test

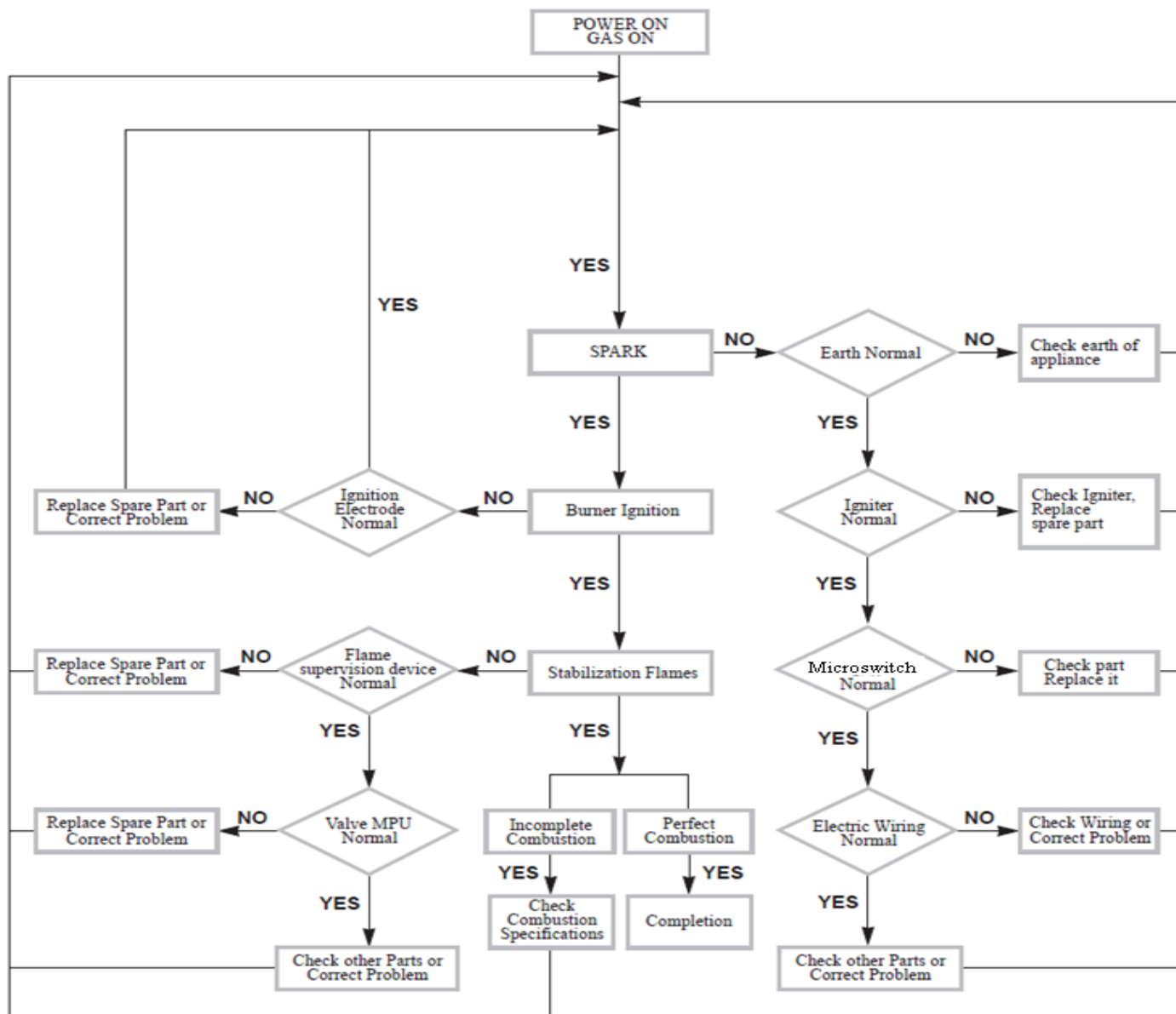


6.5□ Check Point after assemble

- a. At completion of the appliance repair, assemble after removing tools inside of the appliance.
- b. You must conduct a leakage test after repairing.(refer to the leakage test on page 18)
- c. If ignition does not work, repeat ignition several times so that you can remove remaining air in the gas connection
- d. Check combustion status by conducting aging

7.Trouble shooting

7.1. Trouble shooting



7.2. Problems and Corrective Measures

Problem	Probable cause	Solution
Can't ignite	No Spark.	Check battery power.
	The burner cap is badly assembled.	Assemble the cap correctly.
	The gas supply is closed.	Open the gas supply completely.
Difficult to ignite	The gas supply is not completely open.	Open the gas supply completely.
	The burner cap is badly assembled.	Assemble the cap correctly.
	The ignition plug is contaminated with alien substance.	Wipe alien substance with a dry cloth.
	The burners are wet.	Dry the burners cap carefully.
	The holes in the flame spreader are clogged.	Clean the flame spreader.
Noise made when combusted and ignited	The burner cap is badly assembled.	Assemble the burner cap correctly.
Flame goes off when in use.	The flame supervision device is contaminated with alien substance.	Clean the flame supervision device.
	Product being cooked has boiled over and extinguished the flame.	Turn the knob fully clockwise to gas off position. Wait for 1 minute before re-igniting. Refer to pg10 & 11 for igniting & turning off procedure.
	A strong draught may have blown the flame out.	
Yellow Flame	The holes in the flame spreader are clogged.	Clean the flame spreader.
	Different gas is used.	Check the gas used.
Unstable Flame	The burner cap is badly assembled.	Assemble the burner cap correctly.
Gas Smell	Gas leakage	Stop using the product and close the main valve. Open the window to ventilate. Contact our service centre by using a phone outside.

Note!

- 1 Check the ground wire before the fault is detected;
- 2 Enough attention must be given to high voltage circuit.

7.3 Maintenance and test

Air leakage test

Tool: Air leakage device, soapy water

1. Professional repair procedures

- a) Connect product to the air leakage device and set the test pressure at 15kPa;
 - b) The plug valve shall be kept closed and the gas pipelines may be tested its leakage through air leakage device;
 - c) All plug valves shall be kept open, block the injectors at the burner, the flame-out protection valve shall be kept open and the gas pipelines may be tested their leakage through air leakage device.
- The maximum leakage must be controlled below 0.5mL/min as required in the test.



2. Simple procedures:

- a) Connect the product to the gas source being used;
- b) Smear soapy water at each joint of the product gas pipelines, the plug valve is kept closed, check visually each joint for air bubbles;
- c) Smear soapy water at each joint of product gas pipelines, all plug valves shall be kept open, block the injectors at the burner, the flame-out protection device valve shall be kept open and check visually for any air bubbles.

No air bubbles shall be detected at each joint as required in the test.

Note: Any component of the product gas pipelines shall not be disassembled or replaced only if the air leakage is proved in conformity with relevant requirements.